

Ziyang Liu 刘子阳

Singapore | +65 8591 7024 | ziyang.liu.r@outlook.com | github.com/ziyangliu-666 | linkedin.com/in/ziyangliu666

EDUCATION

National University of Singapore

M.Tech. in Software Engineering

Aug 2026 – Present

Singapore

University of Electronic Science and Technology of China

B.Eng. in Software Engineering

Sep 2020 – Jun 2024

Chengdu, China

EXPERIENCE

SmartX

Virtualization R&D Engineer

Jul 2024 – Sep 2025

Chengdu, China

V2V OS (1.6.0 to 1.7.2): led the design of the product that moves guests from VMware onto SMTX OS, and between SMTX OS clusters. Snapshot transfer over VMware CBT, a Python control plane, a C core shipped as a shared library. Now on 10,000+ production VMs, including core financial workloads.

Transfer throughput: replaced the serial read-then-write path with async chunked reads and writes across a shared queue, 8 threads per side. Sustained throughput went from 70 MB/s to 290 MB/s.

VDDK version isolation: moved migration tasks from threads in one process to a process each, one VDDK build per process. Opened up far more vSphere releases.

Platform coverage: moved driver injection's disk attach from NFS to iSCSI after NFS mounts hung guests. Adapted the pipeline to SMTX ELF disaggregated compute-storage clusters.

VMTools: rebuilt the in-guest agent, which ran operations as shell scripts through QGA guest-exec and leaked zombies on several guest OSes. Native QGA commands and new resource probes cut latency from 3 s to under 50 ms, and yearly escalations from 40 to under 5.

Filesystem-consistent snapshots: made snapshots filesystem-consistent, not only crash-consistent. The control plane runs fsfreeze in the guest through VMTools before the snapshot, and thaws after.

Cross-cluster live migration: helped build live migration between SMTX OS clusters, and wrote the data check that a libvirt hook fires once it lands. It snapshots both disks through the storage API, then hashes and compares them block by block. Each side holds its compare order in a sequence queue, and the check never blocks switchover.

Guest kernel defects: found with strace a UOS kernel that creates pidfds but never reaps them, hanging customer VMs. The defect went out as a financial-sector advisory and the fix drew a written commendation. Separately root-caused old kernels not re-polling the QGA serial fd, which silently truncated host-to-guest JSON.

SmartX

Virtualization R&D Intern

Oct 2022 – Jul 2024

Chengdu, China

V2V OS (1.2.0 to 1.6.0): built the migration service as a virtual appliance, on an openEuler base for government procurement. Rebuilt its kernel for the feature bits newer guest filesystems need.

Incremental sync: migration used to copy a VM's disk once, then shut the VM down and copy everything that had changed meanwhile. Made that catch-up repeat while the VM keeps running, using VMware CBT to find the changed blocks. The final pass is left with very little to move, and downtime falls with it.

Packaging and release: wrote the rpm.spec packaging for the V2V rpm and its kernel. Built the CI/CD that assembles both into a bootable ISO, and a Packer pipeline for the vSphere OVF image.

Driver injection: repacked guest initramfs on Linux. On Windows, replayed registry deltas captured with procmon around a virtio-win install, across MBR and UEFI. Snapshotted each disk first, with an IDE fallback on failure.

Guest coverage: carried injection and VMTools across nine Linux distributions, among them openEuler, UOS and Kylin, plus Windows Server 2012 to 2025.

ByteDance (Feishu)

Systems Integration Development Intern

Jun 2022 – Sep 2022

Chengdu, China

Workflow integration: built Go and Gin services connecting recruiting workflows to external OA systems, with MongoDB holding sync and retry state.

API hardening: added authentication, rate limiting and a regression suite across the integration endpoints.

PROJECT

Exfer
Full-stack Engineer

May 2026 – Aug 2026
[exfer.info](#)

The network: co-built a proof-of-work chain for machine-to-machine payments with one other developer, its transaction fee fixed in the protocol. Live 90+ days, 550+ nodes in 28 countries, 180+ miners, 1.36 MH/s.

Agent harness: built the in-wallet agent as a headless core: one loop for desktop, mobile and CI against a scripted model. Published the wallet's tools as an MCP server on PyPI, about 40 across 14 modules. Classed each call silent, consent-gated or background, with the gate in the harness, not the tool. Added a spawn tool for read-only research sub-agents. Kept third-party MCP servers read-only, handing payment requests to the native gated transfer.

Chain internals (Rust, [upstream](#)): landed 28 pull requests across consensus, storage and P2P. Persisted the UTXO set to redb, added a reverse-spend index, and fixed a double-open in the atomic reorg commit. Cleared 8 bugs behind an inflated orphan rate, and cut cold-start bootstrap from ~37 minutes to ~1 minute.

Atomic swap: wrote SHA-256 HTLCs on both chains rather than keccak, one preimage for both legs. Ordered the locks user-first on the long timeout. 938 have settled on mainnet.

Wallets: built the [desktop](#) and [mobile](#) apps, both embedding the signing daemon. Behind them, split signing into its own service and wrote a reorg-aware indexer covering every address and HTLC.

RESEARCH

HKUST (Guangzhou)
Researcher, LLM systems: agent memory, decoding, interpretability

Sep 2025 – Aug 2026
Guangzhou, China

Under review

Y. Shi, Z. Liu, K. Luo, Y. Luo, N. Tang. A Living In-the-Wild Benchmark for Measuring the Static-Benchmark Gap in AI-Generated Video Detection. NeurIPS 2026 Datasets and Benchmarks Track submission. VidTide refreshes every month from real platform video, and seven detectors lose 31 AUROC points on average against it.

Y. Shi, K. Luo, Z. Liu, Y. Luo, N. Tang. Neuro-Symbolic Forensic Reasoning for Open-Generator AI-Generated Video Detection. NeurIPS 2026 submission. FOVEA needs no task training: a VLM plans the forensic hypotheses, classical CV tools test them, 97.39% accuracy on ten unseen generators.

In preparation

Committed SAE-feature traces: made a hosted API prove it served the model it advertised. Fingerprinted every answer with a sparse-autoencoder feature trace, committed to a Merkle root before any check is requested. Cheating then means failing the check or paying for the real model anyway. Caught all 17 substitutes across three backbones, under 2.1% overhead.

Cooperative memory paging: kept a long chat usable after it outgrows the context window. Checked first whether the model signals a missing fact through token-level uncertainty. It does not, and answers more confidently without it. Left an ~8-token keyword bookmark in place of each dropped section, plus a recall() tool for the full text. Beat five methods on LoCoMo, four LLM judges agreeing at $p = 0.017$.

Copy-as-decode: cut the cost of re-emitting unchanged text when an LLM edits a file. Had it name the input lines to copy, spliced into the KV cache by one parallel prefill. Speculative decoding, except the draft is the input document and a grammar guarantees acceptance. Copied text is 74 to 98% of an edit, bounding the speedup at $13\times$.

SKILLS

Languages	Python, Rust, Go, C, TypeScript, Shell
LLM	Model Context Protocol, agent tool design, context and memory engineering, LLM-as-judge
Systems	QEMU/KVM, libvirt, Virtio, QEMU Guest Agent, V2V migration, Linux kernel internals
Distributed	UTXO ledgers, HTLC atomic swaps, P2P sync and reorg handling, JSON-RPC, SSE
Tooling	gdb, perf, strace, Docker, GitHub Actions, Jenkins, Git, RPM/DEB packaging
Data	MySQL, MongoDB, Redis, SQLite

HONOURS

National Third Prize, China Collegiate Computer Design Competition (2022) | UESTC Scholarship (2020–21)